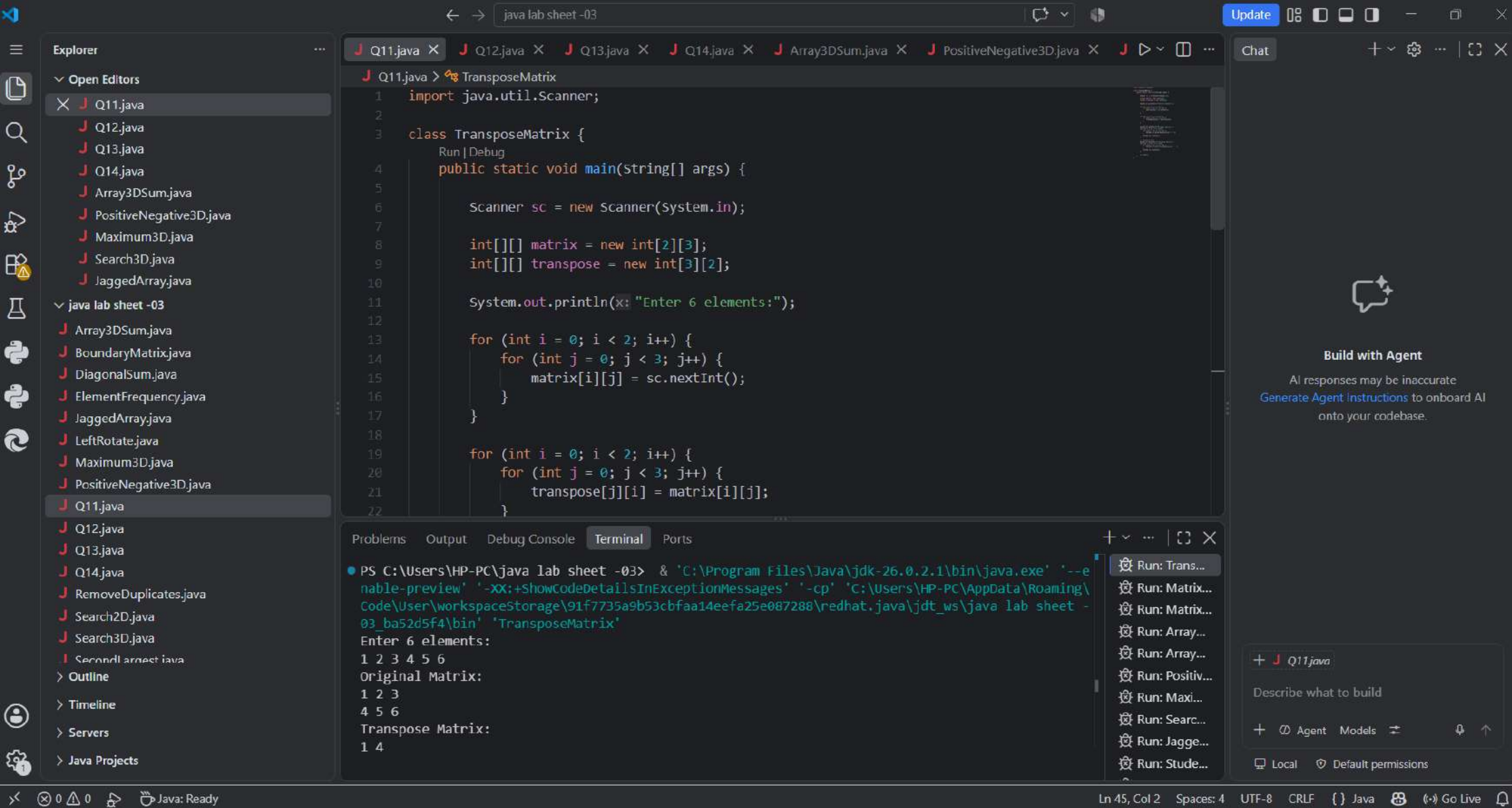
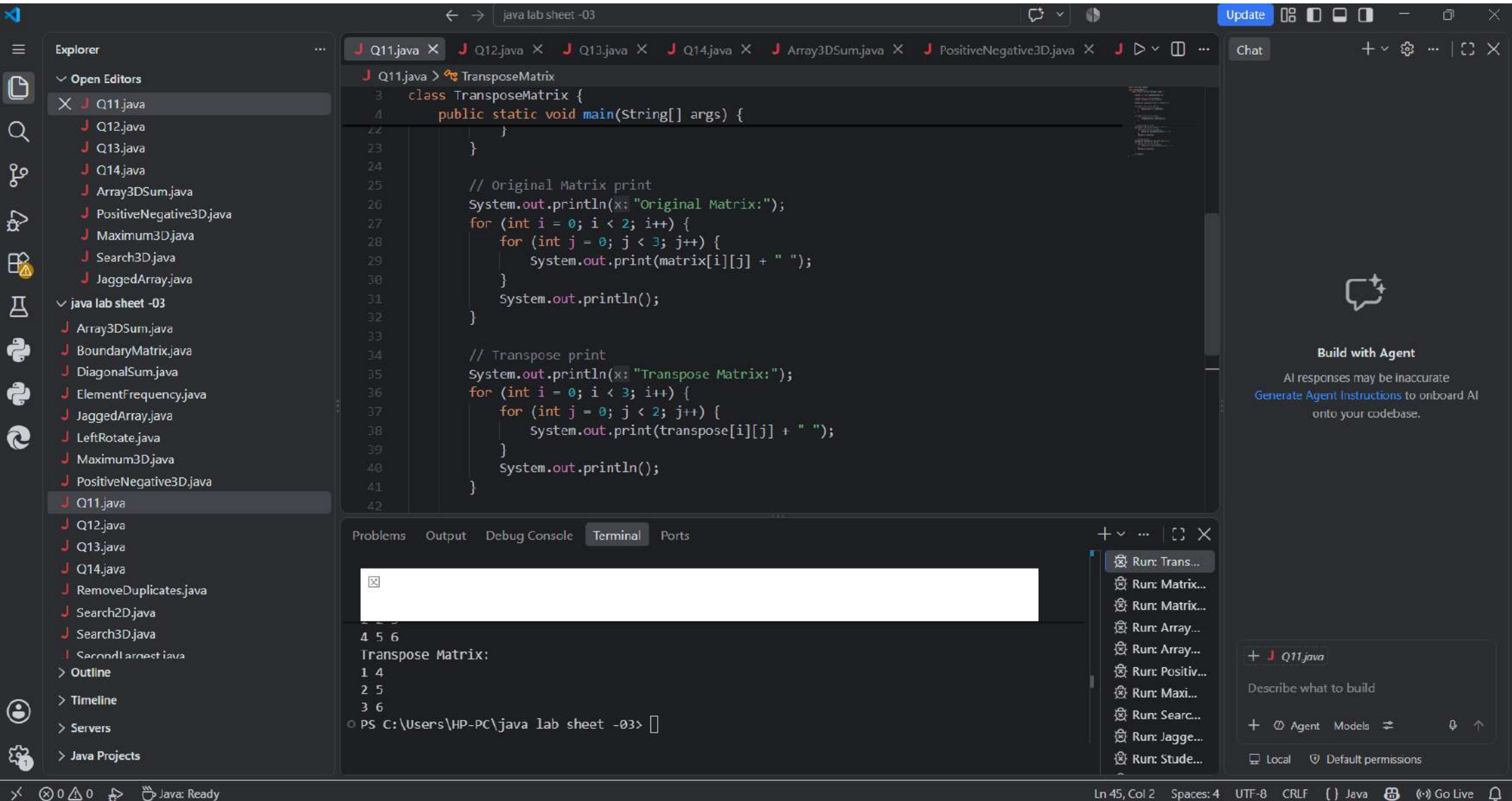






Explorer

Open Editors

Q11.java

Q12.java

Q13.java

Q14.java

Array3DSum.java

PositiveNegative3D.java

Maximum3D.java

Search3D.java

JaggedArray.java

java lab sheet -03

Array3DSum.java

BoundaryMatrix.java

DiagonalSum.java

ElementFrequency.java

JaggedArray.java

LeftRotate.java

Maximum3D.java

PositiveNegative3D.java

Q11.java

Q12.java

Q13.java

Q14.java

RemoveDuplicates.java

Search2D.java

Search3D.java

Second largest java

Outline

Timeline

Servers

Java Projects

Q11.java

Q12.java

Q13.java

Q14.java

Array3DSum.java

PositiveNegative3D.java

Q12.java

MatrixAddition

main(String[])

```
1 import java.util.Scanner;
2
3 class MatrixAddition {
4     Run | Debug
5     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8
9         int[][] a = new int[2][2];
10        int[][] b = new int[2][2];
11        int[][] sum = new int[2][2];
12
13        System.out.println(x: "Enter elements of first matrix:");
14        for (int i = 0; i < 2; i++) {
15            for (int j = 0; j < 2; j++) {
16                a[i][j] = sc.nextInt();
17            }
18        }
19
20        System.out.println(x: "Enter elements of second matrix:");
21        for (int i = 0; i < 2; i++) {
22            for (int j = 0; j < 2; j++) {
```

Problems

Output

Debug Console

Terminal

Ports

```
PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\workspaceStorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'MatrixAddition'
Enter elements of first matrix:
1 2 3 4
Enter elements of second matrix:
5 6 7 8
Sum of matrices:
6 8
10 12
```

Run: Trans...

Run...

Run: Matrix...

Run: Array...

Run: Array...

Run: Positiv...

Run: Maxi...

Run: Searc...

Run: Jagge...

Run: Stude...

+ Q12.java

Describe what to build

+ Agent

Models

Local

Default permissions

Chat

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

Ln 39, Col 20

Spaces: 4

UTF-8

CRLF

{ } Java

Go Live

☰

Explorer

▼ Open Editors

Q11.java

Q12.java

Q13.java

Q14.java

Array3DSum.java

PositiveNegative3D.java

Maximum3D.java

Search3D.java

JaggedArray.java

▼ java lab sheet -03

Array3DSum.java

BoundaryMatrix.java

DiagonalSum.java

ElementFrequency.java

JaggedArray.java

LeftRotate.java

Maximum3D.java

PositiveNegative3D.java

Q11.java

Q12.java

Q13.java

Q14.java

RemoveDuplicates.java

Search2D.java

Search3D.java

SecondLargest.java

> Outline

> Timeline

> Servers

> Java Projects

← → java lab sheet -03

Q11.java Q12.java Q13.java Q14.java Array3DSum.java PositiveNegative3D.java

Q12.java > MatrixAddition > main(String[])

3 class MatrixAddition {

4 public static void main(String[] args) {

23 }

24 }

25 }

26 for (int i = 0; i < 2; i++) {

27 for (int j = 0; j < 2; j++) {

28 sum[i][j] = a[i][j] + b[i][j];

29 }

30 }

31 }

32 System.out.println("Sum of matrices:");

33 for (int i = 0; i < 2; i++) {

34 for (int j = 0; j < 2; j++) {

35 System.out.print(sum[i][j] + " ");

36 }

37 System.out.println();

38 }

39 sc.close();

40 }

41 }

Problems

Output

Debug Console

Terminal

Ports

PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '-e review' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\UspaceStorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_bas2d5f4\bi rixAddition'

5 6 7 8

Sum of matrices:

6 8

10 12

PS C:\Users\HP-PC\java lab sheet -03>

+ - ... | {} X

Run: Trans...

Run: ...

Run: Matrix...

Run: Array...

Run: Array...

Run: Positiv...

Run: Maxi...

Run: Searc...

Run: Jagge...

Run: Stude...

+ Q12.java

Describe what to build

+ Agent Models

Local Default permissions

Update

Chat

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

Ln 39, Col 20 Spaces: 4 UTF-8 CRLF {} Java Go Live

Explorer

Open Editors

Q11.java

Q12.java

Q13.java

Q14.java

Array3DSum.java

PositiveNegative3D.java

Maximum3D.java

Search3D.java

JaggedArray.java

java lab sheet -03

Array3DSum.java

BoundaryMatrix.java

DiagonalSum.java

ElementFrequency.java

JaggedArray.java

LeftRotate.java

Maximum3D.java

PositiveNegative3D.java

Q11.java

Q12.java

Q13.java

Q14.java

RemoveDuplicates.java

Search2D.java

Search3D.java

SecondLargest.java

Outline

Timeline

Servers

Java Projects

Q11.java

Q12.java

Q13.java

Q14.java

Array3DSum.java

PositiveNegative3D.java

Run

Stop

Debug

Test

Q13.java > MatrixMultiplication > main(String[])

```
3 class MatrixMultiplication {
4     public static void main(String[] args) {
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25         for (int i = 0; i < 3; i++) {
26             for (int j = 0; j < 3; j++) {
27                 for (int k = 0; k < 3; k++) {
28                     result[i][j] += a[i][k] * b[k][j];
29                 }
30             }
31         }
32
33         System.out.println(x: "Result of multiplication:");
34         for (int i = 0; i < 3; i++) {
35             for (int j = 0; j < 3; j++) {
36                 System.out.print(result[i][j] + " ");
37             }
38             System.out.println();
39         }
40         sc.close();
41     }
42 }
```

Problems

Output

Debug Console

Terminal

Ports

```
PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-p
review' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\work
spaceStorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'Mat
rixMultiplication'
3 2 1
Result of multiplication:
30 24 18
84 69 54
138 114 90
PS C:\Users\HP-PC\java lab sheet -03> 
```

Run: Trans...

Run: Matrix...

Run: ...

Run: Array...

Run: Array...

Run: Positiv...

Run: Maxi...

Run: Searc...

Run: Jagge...

Run: Stude...

Q13.java

Describe what to build

+ Agent Models

Local Default permissions

Ln 41, Col 6 Spaces: 4 UTF-8 CRLF {} Java Go Live

Q11.java

Q12.java

Q13.java

Q14.java

Array3DSum.java

PositiveNegative3D.java

Maximum3D.java

Search3D.java

JaggedArray.java

java lab sheet -03

Array3DSum.java

BoundaryMatrix.java

DiagonalSum.java

ElementFrequency.java

JaggedArray.java

LeftRotate.java

Maximum3D.java

PositiveNegative3D.java

Q11.java

Q12.java

Q13.java

Q14.java

RemoveDuplicates.java

Search2D.java

Search3D.java

SecondLargest.java

Outline

Timeline

Servers

Java Projects

Q14.java

Array3DDisplay

main(String[] args)

import java.util.Scanner;

class Array3DDisplay {

public static void main(String[] args) {

Scanner sc = new Scanner(System.in);

int[][][] arr = new int[2][2][2];

System.out.println(x: "Enter 8 elements:");

for (int i = 0; i < 2; i++) {

for (int j = 0; j < 2; j++) {

for (int k = 0; k < 2; k++) {

arr[i][j][k] = sc.nextInt();

}

}

}

System.out.println(x: "Elements of 3-D array:");

for (int i = 0; i < 2; i++) {

System.out.println("Layer " + (i + 1) + ":");

}

Problems

Output

Debug Console

Terminal

Ports

PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-p
review' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\work
spaceStorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'Arr
ay3DDisplay'
Enter 8 elements:
1 2 3 4 5 6 7 8
Elements of 3-D array:
Layer 1:
1 2
3 4
Layer 2:

Run: Trans...

Run: Matrix...

Run: Matrix...

Run: ...

Run: Array...

Run: Positiv...

Run: Maxi...

Run: Searc...

Run: Jagge...

Run: Stude...

Update

Chat

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

+ Q14.java

Describe what to build

+ Agent Models

Local Default permissions

Ln 30, Col 24 Spaces: 4 UTF-8 CRLF {} Java Go Live

Explorer

Open Editors

Q11.java

Q12.java

Q13.java

Q14.java

Array3DSum.java

PositiveNegative3D.java

Maximum3D.java

Search3D.java

JaggedArray.java

java lab sheet -03

Array3DSum.java

BoundaryMatrix.java

DiagonalSum.java

ElementFrequency.java

JaggedArray.java

LeftRotate.java

Maximum3D.java

PositiveNegative3D.java

Q11.java

Q12.java

Q13.java

Q14.java

RemoveDuplicates.java

Search2D.java

Search3D.java

SecondLargest.java

Outline

Timeline

Servers

Java Projects

Q11.java

Q12.java

Q13.java

Q14.java

Array3DSum.java

PositiveNegative3D.java

Run

Q14.java > Array3DDisplay > main(String[])

```
3 class Array3DDisplay {
4     public static void main(String[] args) {
17
18
19         System.out.println(x: "Elements of 3-D array:");
20
21         for (int i = 0; i < 2; i++) {
22             System.out.println("Layer " + (i + 1) + ":");
23
24             for (int j = 0; j < 2; j++) {
25                 for (int k = 0; k < 2; k++) {
26                     System.out.print(arr[i][j][k] + " ");
27                 }
28                 System.out.println();
29             }
30             sc.close();
31         }
32     }
33 }
```

Problems

Output

Debug Console

Terminal

Ports

```
PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-p
review' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\work
spaceStorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'Arr
ay3DDisplay'
1 2
3 4
Layer 2:
5 6
7 8
PS C:\Users\HP-PC\java lab sheet -03>
```

Run: Trans...

Run: Matrix...

Run: Matrix...

Run: ...

Run: Array...

Run: Positiv...

Run: Maxi...

Run: Searc...

Run: Jagge...

Run: Stude...

Chat

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

+ Q14.java

Describe what to build

+ Agent Models

Local Default: permissions

Ln 30, Col 24 Spaces: 4 UTF-8 CRLF {} Java Go Live

Explorer

Open Editors

Q11.java

Q12.java

Q13.java

Q14.java

Array3DSum.java

PositiveNegative3D.java

Maximum3D.java

Search3D.java

JaggedArray.java

java lab sheet -03

Array3DSum.java

BoundaryMatrix.java

DiagonalSum.java

ElementFrequency.java

JaggedArray.java

LeftRotate.java

Maximum3D.java

PositiveNegative3D.java

Q11.java

Q12.java

Q13.java

Q14.java

RemoveDuplicates.java

Search2D.java

Search3D.java

SecondLargest.java

Outline

Timeline

Servers

Java Projects

java lab sheet -03

Array3DSum.java

Q11.java

Q12.java

Q13.java

Q14.java

PositiveNegative3D.java

Run | Debug

```
1 import java.util.Scanner;
2
3 public class Array3DSum {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         int[][][] arr = new int[2][2][2];
9         int sum = 0;
10
11         System.out.println("Enter 8 elements:");
12
13         for (int i = 0; i < 2; i++) {
14             for (int j = 0; j < 2; j++) {
15                 for (int k = 0; k < 2; k++) {
16                     arr[i][j][k] = sc.nextInt();
17                     sum = sum + arr[i][j][k];
18                 }
19             }
20         }
21
22         System.out.println("Sum of all elements = " + sum);
23     }
24 }
```

Problems

Output

Debug Console

Terminal

Ports

```
PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-p
review' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\work
spaceStorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'Arr
ay3DSum'
Enter 8 elements:
1
2
3
4
5
6
```

Run: Trans...

Run: Matrix...

Run: Matrix...

Run: Array...

Run: ...

Run: Positiv...

Run: Maxi...

Run: Search...

Run: Jagge...

Run: Stude...

Chat

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

Array3DSum.java

Describe what to build

Agent

Models

Local

Default permissions

0

0

Java: Ready

Ln 26, Col 2

Spaces: 4

UTF-8

CRLF

{ } Java

Go Live

Explorer

Open Editors

Q11.java

Q12.java

Q13.java

Q14.java

Array3DSum.java

PositiveNegative3D.java

Maximum3D.java

Search3D.java

JaggedArray.java

java lab sheet -03

Array3DSum.java

BoundaryMatrix.java

DiagonalSum.java

ElementFrequency.java

JaggedArray.java

LeftRotate.java

Maximum3D.java

PositiveNegative3D.java

Q11.java

Q12.java

Q13.java

Q14.java

RemoveDuplicates.java

Search2D.java

Search3D.java

SecondLargest.java

Outline

Timeline

Servers

Java Projects

java lab sheet -03

Array3DSum.java

```
3 public class Array3DSum {
4     public static void main(String[] args) {
13         for (int i = 0; i < 2; i++) {
14             for (int j = 0; j < 2; j++) {
15                 for (int k = 0; k < 2; k++) {
16                     arr[i][j][k] = sc.nextInt();
17                     sum = sum + arr[i][j][k];
18                 }
19             }
20         }
21
22         System.out.println("Sum of all elements = " + sum);
23
24         sc.close();
25     }
26 }
```

Problems

Output

Debug Console

Terminal

Ports

PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-p
review' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\work
spaceStorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'Arr
ay3DSum'
4
5
6
7
8
Sum of all elements = 36

Run: Trans...

Run: Matrix...

Run: Matrix...

Run: Array...

Run: ...

Run: Positiv...

Run: Maxi...

Run: Search...

Run: Jagge...

Run: Stude...

Chat

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

+

Array3DSum.java

Describe what to build

+

Agent

Models

Local

Default permissions

Ln 26, Col 2 Spaces: 4 UTF-8 CRLF [] Java Go Live

☰

Open Editors

Q11.java

Q12.java

Q13.java

Q14.java

Array3DSum.java

PositiveNegative3D.java

Maximum3D.java

Search3D.java

JaggedArray.java

java lab sheet -03

Array3DSum.java

BoundaryMatrix.java

DiagonalSum.java

ElementFrequency.java

JaggedArray.java

LeftRotate.java

Maximum3D.java

PositiveNegative3D.java

Q11.java

Q12.java

Q13.java

Q14.java

RemoveDuplicates.java

Search2D.java

Search3D.java

SecondLargest.java

Outline

Timeline

Servers

Java Projects

java lab sheet -03

Q11.java Q12.java Q13.java Q14.java Array3DSum.java PositiveNegative3D.java

PositiveNegative3D.java

```
1 import java.util.Scanner;
2
3 public class PositiveNegative3D {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         int[][][] arr = new int[2][2][2];
9
10        int positive = 0;
11        int negative = 0;
12
13        System.out.println("Enter 8 elements:");
14
15        for (int i = 0; i < 2; i++) {
16            for (int j = 0; j < 2; j++) {
17                for (int k = 0; k < 2; k++) {
18
19                    arr[i][j][k] = sc.nextInt();
20
21                    if (arr[i][j][k] > 0) {
22                        positive++;
```

Problems

Output

Debug Console

Terminal

Ports

PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\workspaceStorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'PositiveNegative3D'

Enter 8 elements:

10 -5 7 -2 8 4 -3

9

Positive numbers = 5

Negative numbers = 3

PS C:\Users\HP-PC\java lab sheet -03>

Run: Trans...

Run: Matrix...

Run: Matrix...

Run: Array...

Run: Array...

Run: ...

Run: Maxi...

Run: Searc...

Run: Jagge...

Run: Stude...

+ PositiveNegative3D.java

Describe what to build

+ Agent Models

Local Default permissions

Ln 35, Col 2 Spaces: 4 UTF-8 CRLF {} Java Go Live

☰

Open Editors

Q11.java

Q12.java

Q13.java

Q14.java

Array3DSum.java

PositiveNegative3D.java

Maximum3D.java

Search3D.java

JaggedArray.java

java lab sheet -03

Array3DSum.java

BoundaryMatrix.java

DiagonalSum.java

ElementFrequency.java

JaggedArray.java

LeftRotate.java

Maximum3D.java

PositiveNegative3D.java

Q11.java

Q12.java

Q13.java

Q14.java

RemoveDuplicates.java

Search2D.java

Search3D.java

SecondLargest.java

Outline

Timeline

Servers

Java Projects

java lab sheet -03

PositiveNegative3D.java

```
public class PositiveNegative3D {
    public static void main(String[] args) {
        int arr[][][] = new int[3][3][3];

        if (arr[i][j][k] > 0) {
            positive++;
        } else if (arr[i][j][k] < 0) {
            negative++;
        }
    }
}

System.out.println("Positive numbers = " + positive);
System.out.println("Negative numbers = " + negative);

sc.close();
}
```

Problems

Output

Debug Console

Terminal

Ports

```
PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-p
review' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\work
spaceStorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'Pos
itiveNegative3D'
Enter 8 elements:
10 -5 7 -2 8 4 -3
9
Positive numbers = 5
Negative numbers = 3
PS C:\Users\HP-PC\java lab sheet -03>
```

Run: Trans...

Run: Matrix...

Run: Matrix...

Run: Array...

Run: Array...

Run: Run... [icon] [icon]

Run: Maxi...

Run: Search...

Run: Jagge...

Run: Stude...

Chat

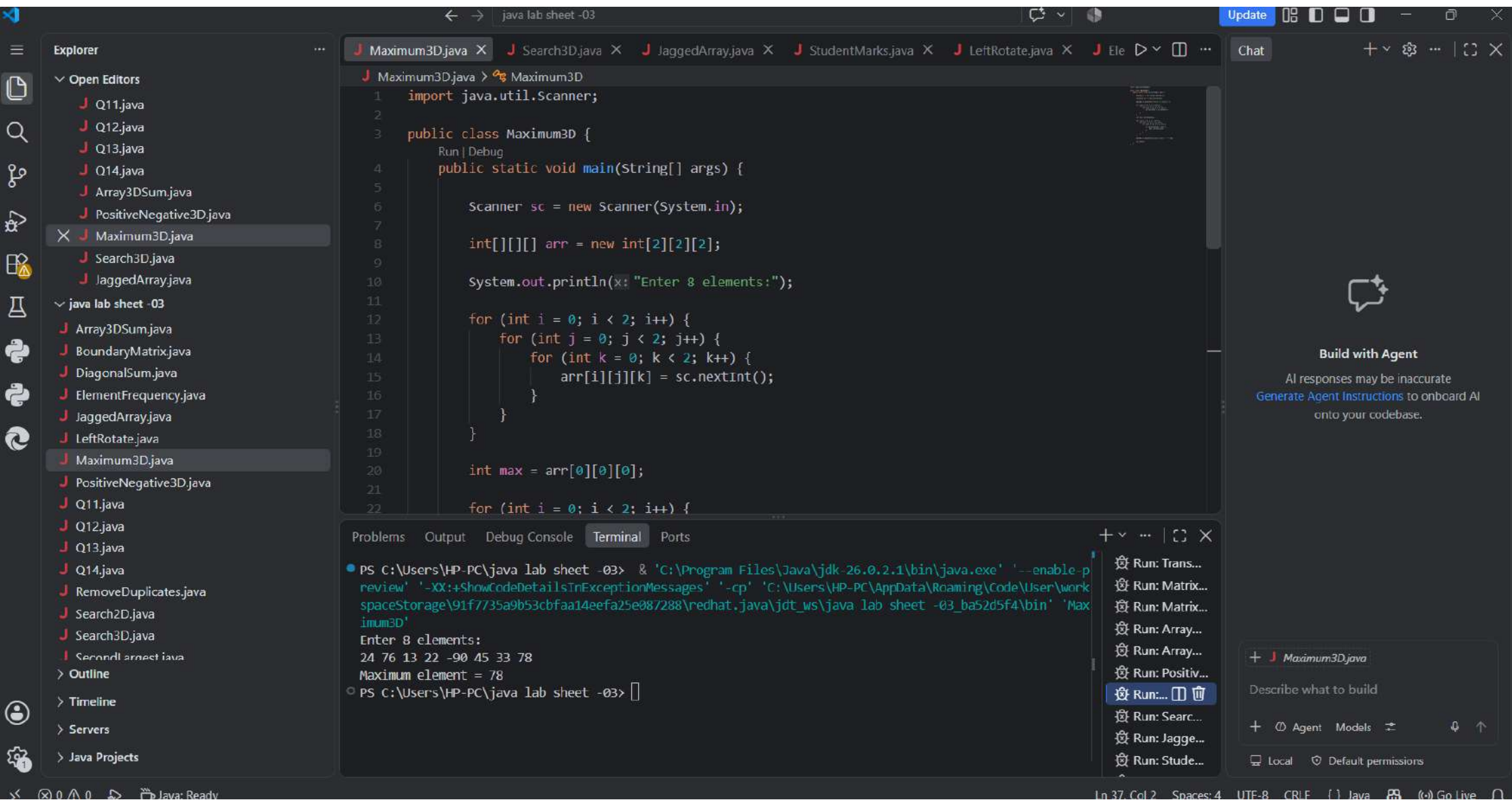
Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI

onto your codebase.

Ln 35, Col 2 Spaces: 4 UTF-8 CRLF {} Java [icon] Go Live [icon]



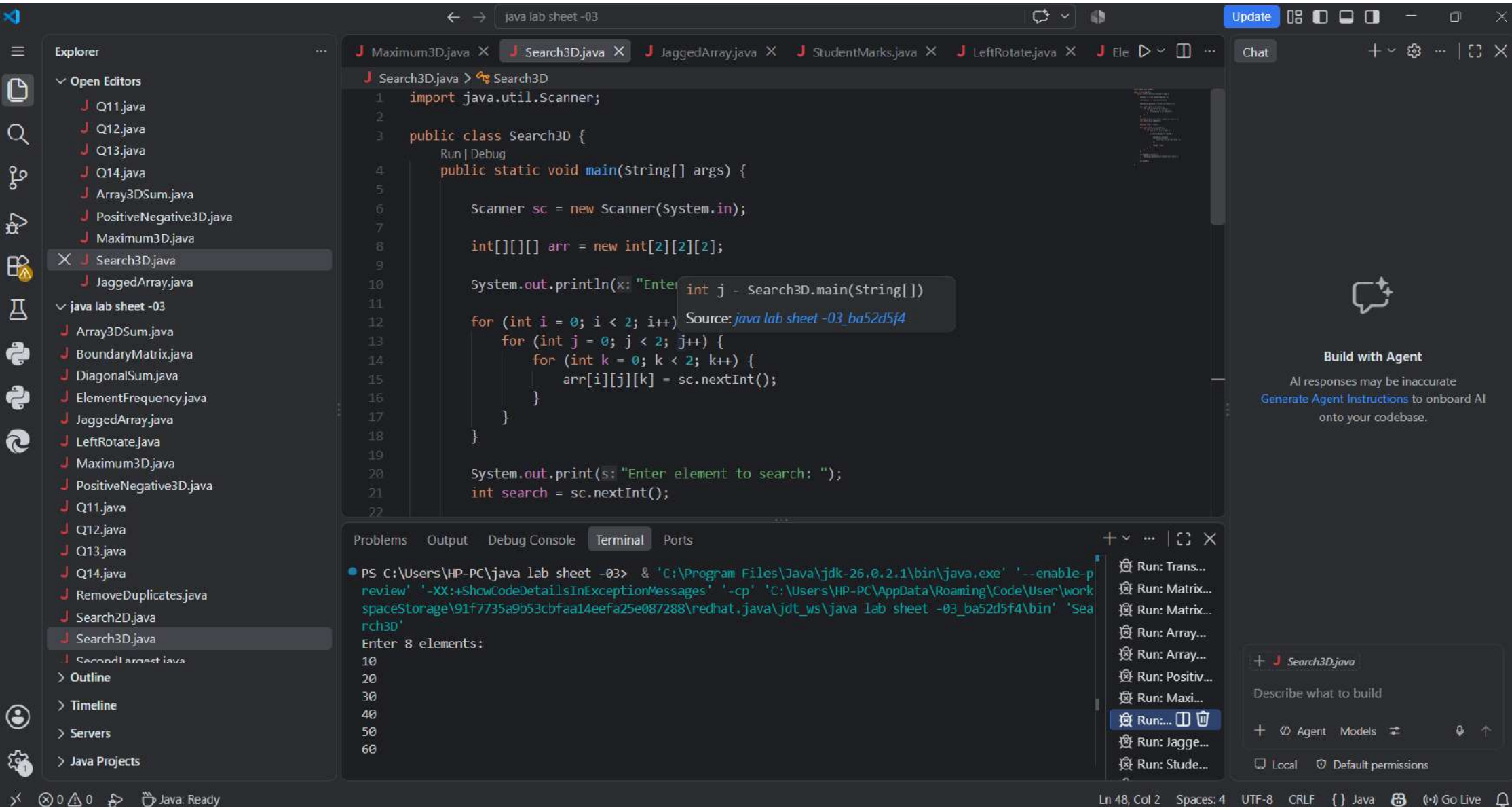
The screenshot displays an IDE with a dark theme. On the left, the Explorer panel shows a project named 'java lab sheet -03' containing various Java files. The file 'Maximum3D.java' is selected and open in the editor. The code in the editor is as follows:

```
3 public class Maximum3D {
4     public static void main(String[] args) {
5
6         int max = arr[0][0][0];
7
8         for (int i = 0; i < 2; i++) {
9             for (int j = 0; j < 2; j++) {
10                 for (int k = 0; k < 2; k++) {
11                     if (arr[i][j][k] > max) {
12                         max = arr[i][j][k];
13                     }
14                 }
15             }
16         }
17
18         System.out.println("Maximum element = " + max);
19
20         sc.close();
21     }
22 }
```

Below the editor, the Terminal panel is active, showing the command prompt output:

```
PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-p
review' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\work
spaceStorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'Max
imum3D'
Enter 8 elements:
24 76 13 22 -90 45 33 78
Maximum element = 78
PS C:\Users\HP-PC\java lab sheet -03>
```

On the right side of the IDE, there is a 'Chat' panel with a message icon and the text 'Build with Agent'. Below this, it says 'AI responses may be inaccurate' and 'Generate Agent Instructions to onboard AI onto your codebase.' At the bottom right, there is a 'Run' button with a play icon and a 'Run' button with a play icon and a 'Run' button with a play icon.



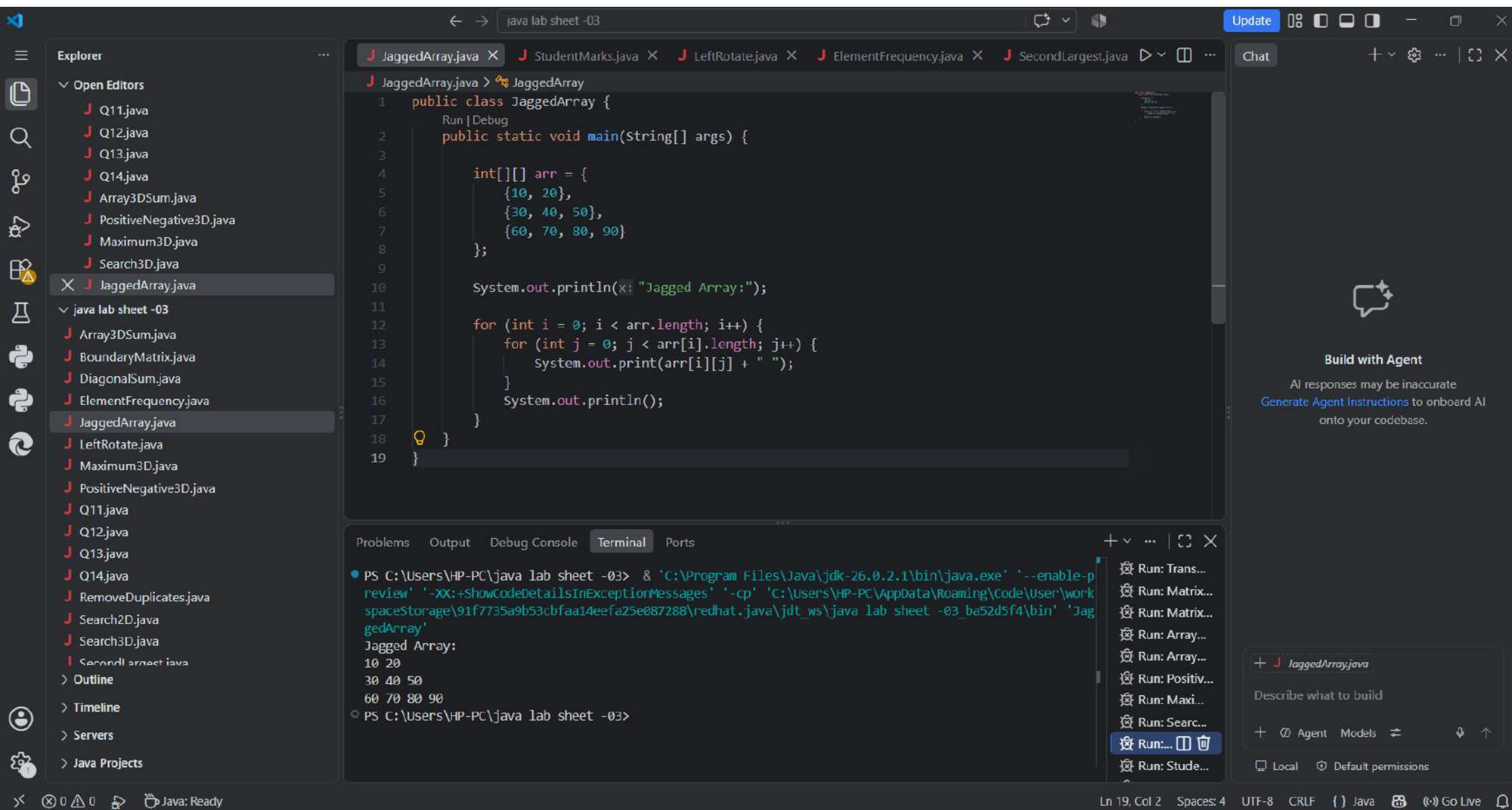
The screenshot displays the IntelliJ IDEA IDE interface. The Explorer panel on the left shows the project structure, including a folder named 'java lab sheet -03' containing several Java files. The 'Search3D.java' file is selected and open in the editor. The code in the editor is as follows:

```
public class Search3D {
    public static void main(String[] args) {
        // ... (code for finding an element in a 3D array) ...
        if (found == false) {
            System.out.println(x: "Element not found.");
        }
        sc.close();
    }
}
```

The terminal at the bottom shows the command to run the program and the output:

```
PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\workspaceStorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin\' 'Search3D'
40
50
60
70
80
Enter element to search: 50
```

The right sidebar shows the 'Build with Agent' section, which includes a chat interface for generating agent instructions.



Explorer

Open Editors

Q12.java

Q13.java

Q14.java

Array3DSum.java

PositiveNegative3D.java

Maximum3D.java

Search3D.java

JaggedArray.java

StudentMarks.java

java lab sheet -03

LeftRotate.java

Maximum3D.java

PositiveNegative3D.java

Q11.java

Q12.java

Q13.java

Q14.java

RemoveDuplicates.java

Search2D.java

Search3D.java

SecondLargest.java

SortJaggedArray.java

SparseMatrix.java

StudentMarks.java

TriangularMatrix.java

Outline

Timeline

Servers

Java Projects

JaggedArray.java

StudentMarks.java

LeftRotate.java

ElementFrequency.java

SecondLargest.java

StudentMarks.java

StudentMarks

1 public class StudentMarks {

2 Run | Debug

3 public static void main(String[] args) {

4 int[][] marks = {

5 {80, 75},

6 {90, 85, 88},

7 {70, 65, 72, 78}

8 };

9

10 for (int i = 0; i < marks.length; i++) {

11 System.out.println("Student " + (i + 1) + " marks:");

12

13 for (int j = 0; j < marks[i].length; j++) {

14 System.out.print(marks[i][j] + " ");

15 }

16

17 System.out.println();

18 }

19 }

20 }

Problems

Output

Debug Console

Terminal

Ports

PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-p

review' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\work

spaceStorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'Stu

dentMarks'

Student 1 marks:

80 75

Student 2 marks:

90 85 88

Student 3 marks:

70 65 72 78

PS C:\Users\HP-PC\java lab sheet -03>

Run: Trans...

Run: Matrix...

Run: Matrix...

Run: Array...

Run: Array...

Run: Positiv...

Run: Maxi...

Run: Searc...

Run: Jagge...

Run: Stude...

Chat

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

Dictation

Speak and it becomes text. Adjust settings or how it's written any time.

+ StudentMarks.java

Describe what to build

+ Agent Models

Local Default: permissions

Ln 20, Col 2 Spaces: 4 UTF-8 CRLF {} Java Go Live

☰

☰

🔍

🔗

🔧

🚨

🧪

📄

🔄

👤

⚙️

Explorer

Open Editors

Q13.java

Q14.java

Array3DSum.java

PositiveNegative3D.java

Maximum3D.java

Search3D.java

JaggedArray.java

StudentMarks.java

LeftRotate.java

java lab sheet -03

Array3DSum.java

BoundaryMatrix.java

DiagonalSum.java

ElementFrequency.java

JaggedArray.java

LeftRotate.java

Maximum3D.java

PositiveNegative3D.java

Q11.java

Q12.java

Q13.java

Q14.java

RemoveDuplicates.java

Search2D.java

Search3D.java

SecondLargest.java

Outline

Timeline

Servers

Java Projects

← → java lab sheet -03

🗨️ 🌐

Update

☰

🖨️

🔍

🔄

🗑️

✖

JaggedArray.java X StudentMarks.java X LeftRotate.java X ElementFrequency.java X SecondLargest.java

LeftRotate.java > LeftRotate

1 public class LeftRotate {
2 Run | Debug
3 public static void main(String[] args) {
4 int[] arr = {10, 20, 30, 40, 50};
5
6 int first = arr[0];
7
8 for (int i = 0; i < arr.length - 1; i++) {
9 arr[i] = arr[i + 1];
10 }
11
12 arr[arr.length - 1] = first;
13
14 system.out.println("Array after left rotation:");
15
16 for (int i = 0; i < arr.length; i++) {
17 system.out.print(arr[i] + " ");
18 }
19 }
20 }

Problems Output Debug Console Terminal Ports

PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-p
review' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\work
spaceStorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'Lef
tRotate'
Array after left rotation:
20 30 40 50 10
PS C:\Users\HP-PC\java lab sheet -03>

Run... 🗑️
Run: Eleme...
Run: Secon...
Run: Remo...
Run: Diago...
Run: Triang...
Run: Boun...
Run: Searc...
Run: SortJa...
Run: Spars...

Chat

+ 🔧 ⋮ 🔄 ✖

🗨️

Build with Agent

AI responses may be inaccurate
[Generate Agent Instructions](#) to onboard AI
onto your codebase.

Dictation

✖

Speak and it becomes text. Adjust
[settings](#) or [how it's written](#) any time.

⋮

+ LeftRotate.java

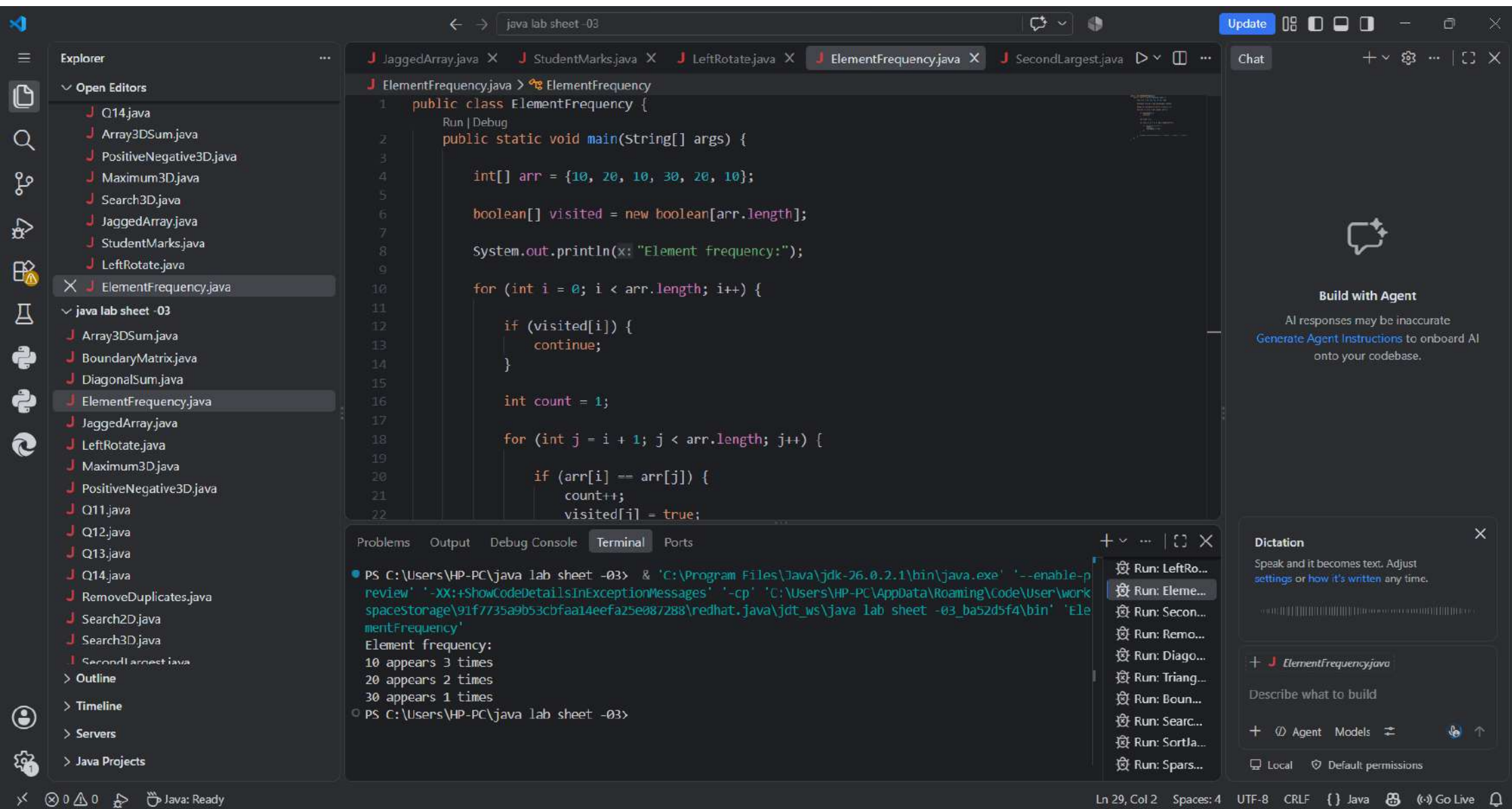
Describe what to build

+ 🗨️ Agent Models 🔧 🔄 📶 ⬆

🗨️ Local 🗨️ Default permissions

🔍 0 0 0 🔄 Java: Ready

Ln 20, Col 2 Spaces: 4 UTF-8 CRLF () Java 🗨️ 🔄 Go Live 🔔



☰

Explorer

▼ Open Editors

Q14.java

Array3DSum.java

PositiveNegative3D.java

Maximum3D.java

Search3D.java

JaggedArray.java

StudentMarks.java

LeftRotate.java

✕ ElementFrequency.java

▼ java lab sheet -03

Array3DSum.java

BoundaryMatrix.java

DiagonalSum.java

ElementFrequency.java

JaggedArray.java

LeftRotate.java

Maximum3D.java

PositiveNegative3D.java

Q11.java

Q12.java

Q13.java

Q14.java

RemoveDuplicates.java

Search2D.java

Search3D.java

SearchAndAnnotate.java

> Outline

> Timeline

> Servers

> Java Projects

← →

java lab sheet -03

Sign In

Search3D.java

JaggedArray.java

StudentMarks.java

LeftRotate.java

ElementFrequency.java

ElementFrequency.java

> ElementFrequency

```
1 public class ElementFrequency {
2     public static void main(String[] args) {
13         continue;
14     }
15
16     int count = 1;
17
18     for (int j = i + 1; j < arr.length; j++) {
19
20         if (arr[i] == arr[j]) {
21             count++;
22             visited[j] = true;
23         }
24     }
25
26     System.out.println(arr[i] + " appears " + count + " times");
27 }
28
29
```

Problems

Output

Debug Console

Terminal

Ports

PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\workspaceStorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'ElementFrequency'

Element frequency:

10 appears 3 times

20 appears 2 times

30 appears 1 times

PS C:\Users\HP-PC\java lab sheet -03>

poweresh...

Run: Eleme...

✕ ElementFrequency.java

Explore and understand your code

+ ⓘ Ask Models

Local

11 29 Col 2 Spaces: 4 UTF-8 CR LF {} Java Go Live

Explorer

Open Editors

Array3DSum.java
PositiveNegative3D.java
Maximum3D.java
Search3D.java
JaggedArray.java
StudentMarks.java
LeftRotate.java
ElementFrequency.java
SecondLargest.java

java lab sheet -03

LeftRotate.java
Maximum3D.java
PositiveNegative3D.java
Q11.java
Q12.java
Q13.java
Q14.java
RemoveDuplicates.java
Search2D.java
Search3D.java
SecondLargest.java
SortJaggedArray.java
SparseMatrix.java
StudentMarks.java
TriangularMatrix.java

Outline
Timeline
Servers
Java Projects

SecondLargest.java

SecondLargest

```
1 public class SecondLargest {  
2     Run | Debug  
3     public static void main(String[] args) {  
4  
5         int[] arr = {10, 50, 30, 80, 40};  
6  
7         int largest = Integer.MIN_VALUE;  
8         int secondLargest = Integer.MIN_VALUE;  
9  
10        for (int i = 0; i < arr.length; i++) {  
11  
12            if (arr[i] > largest) {  
13                secondLargest = largest;  
14                largest = arr[i];  
15            } else if (arr[i] > secondLargest && arr[i] != largest) {  
16                secondLargest = arr[i];  
17            }  
18        }  
19  
20        System.out.println("Largest element = " + largest);  
21        System.out.println("Second largest element = " + secondLargest);  
22    }  
23 }
```

Problems

Output

Debug Console

Terminal

Ports

PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-p
review' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\code\User\work
spaceStorage\91f7735a9b53cbfaa14cefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'Sec
ondLargest'
Largest element = 80
Second largest element = 50
PS C:\Users\HP-PC\java lab sheet -03>

Run: LeftRo...

Run: Eleme...

Run: ...

Run: Remo...

Run: Diago...

Run: Triang...

Run: Boun...

Run: Searc...

Run: SortJa...

Run: Spars...

Chat

Build with Agent

AI responses may be inaccurate
Generate Agent Instructions to onboard AI
onto your codebase.

Dictation

Speak and it becomes text. Adjust
settings or how it's written any time.

+ SecondLargest.java

Describe what to build

+ Agent Models

Local Default permissions

Ln 22, Col 2 Spaces: 4 UTF-8 CRLF {} Java Go Live

The screenshot shows the Visual Studio Code (VS Code) IDE interface. The main editor window displays the file `RemoveDuplicates.java` with the following code:

```
1 public class RemoveDuplicates {
2     public static void main(String[] args) {
3
4         int[] arr = {10, 20, 10, 30, 20, 40, 30};
5
6         int[] result = new int[arr.length];
7         int n = 0;
8
9         for (int i = 0; i < arr.length; i++) {
10
11             boolean duplicate = false;
12
13             for (int j = 0; j < n; j++) {
14
15                 if (arr[i] == result[j]) {
16                     duplicate = true;
17                     break;
18                 }
19             }
20
21             if (!duplicate) {
22                 result[n] = arr[i];
```

The Explorer sidebar on the left shows the project structure, including the file `RemoveDuplicates.java`. The Terminal panel at the bottom shows the command to run the program and the output:

```
PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-p
review' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\work
spaceStorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'Rem
oveDuplicates'
Array after removing duplicates:
10 20 30 40
PS C:\Users\HP-PC\java lab sheet -03>
```

The Chat sidebar on the right is visible, showing a message from the AI agent.

Explorer

Open Editors

Search3D.java

JaggedArray.java

StudentMarks.java

LeftRotate.java

ElementFrequency.java

SecondLargest.java

RemoveDuplicates.java

DiagonalSum.java

TriangularMatrix.java

java lab sheet -03

LeftRotate.java

Maximum3D.java

PositiveNegative3D.java

Q11.java

Q12.java

Q13.java

Q14.java

RemoveDuplicates.java

Search2D.java

Search3D.java

SecondLargest.java

SortJaggedArray.java

SparseMatrix.java

StudentMarks.java

TriangularMatrix.java

Outline

Timeline

Servers

Java Projects

SecondLargest.java

RemoveDuplicates.java

DiagonalSum.java

TriangularMatrix.java

BoundaryMat

TriangularMatrix.java

TriangularMatrix > main(String[])

```
1 public class TriangularMatrix {
2     public static void main(String[] args) {
3
4         int[][] matrix = {
5             {1, 2, 3},
6             {4, 5, 6},
7             {7, 8, 9}
8         };
9
10        System.out.println("Upper Triangular Matrix:");
11
12        for (int i = 0; i < 3; i++) {
13            for (int j = 0; j < 3; j++) {
14
15                if (i <= j) {
16                    System.out.print(matrix[i][j] + " ");
17                } else {
18                    System.out.print("0 ");
19                }
20            }
21            System.out.println();
22        }
23    }
24 }
```

Problems

Output

Debug Console

Terminal

Ports

```
PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-p
review' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\work
spaceStorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'Tri
angularMatrix'
Upper Triangular Matrix:
1 2 3
0 5 6
0 0 9
Lower Triangular Matrix:
1 0 0
4 5 0
```

Run: LeftRo...

Run: Eleme...

Run: Secon...

Run: Remo...

Run: Diago...

Run: Triang...

Run: Boun...

Run: Searc...

Run: SortJa...

Run: Spars...

Chat

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

Dictation

Speak and it becomes text. Adjust settings or how it's written any time.

Listening...

Agent Models

Local Default permissions

Ln 19, Col 18 Spaces: 4 UTF-8 CRLF {} Java Go Live

🔍

📁

🔍

🔗

🔧

📦

🧪

🔗

🔗

🔄

👤

⚙️

Explorer

Open Editors

Search3D.java

JaggedArray.java

StudentMarks.java

LeftRotate.java

ElementFrequency.java

SecondLargest.java

RemoveDuplicates.java

DiagonalSum.java

TriangularMatrix.java

java lab sheet -03

LeftRotate.java

Maximum3D.java

PositiveNegative3D.java

Q11.java

Q12.java

Q13.java

Q14.java

RemoveDuplicates.java

Search2D.java

Search3D.java

SecondLargest.java

SortJaggedArray.java

SparseMatrix.java

StudentMarks.java

TriangularMatrix.java

Outline

Timeline

Servers

Java Projects

java lab sheet -03

SecondLargest.java

RemoveDuplicates.java

DiagonalSum.java

TriangularMatrix.java

BoundaryMat

Chat

TriangularMatrix.java

1 public class TriangularMatrix {

2 public static void main(String[] args) {

18 System.out.print(s: "0 ");

19 }

20 }

21 System.out.println();

22 }

23 }

24 System.out.println(x: "Lower Triangular Matrix:");

25 }

26 for (int i = 0; i < 3; i++) {

27 for (int j = 0; j < 3; j++) {

28 }

29 if (i >= j) {

30 System.out.print(matrix[i][j] + " ");

31 } else {

32 System.out.print(s: "0 ");

33 }

34 }

35 System.out.println();

36 }

37 }

38 }

Problems

Output

Debug Console

Terminal

Ports

PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\workspaceStorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'TriangularMatrix'

0 0 9

Lower Triangular Matrix:

1 0 0

4 5 0

7 8 9

PS C:\Users\HP-PC\java lab sheet -03>

Run: LeftRo...

Run: Eleme...

Run: Secon...

Run: Remo...

Run: Diago...

Run: Triang...

Run: Boun...

Run: Searc...

Run: SortJa...

Run: Spars...

Dictation

Speak and it becomes text. Adjust settings or how it's written any time.

+ TriangularMatrix.java

Listening...

+ Agent Models

Local Default permissions

Ln 19, Col 18

Spaces: 4

UTF-8

CRLF

{ } Java

Go Live

Explorer

Open Editors

JaggedArray.java

StudentMarks.java

LeftRotate.java

ElementFrequency.java

SecondLargest.java

RemoveDuplicates.java

DiagonalSum.java

TriangularMatrix.java

BoundaryMatrix.java

java lab sheet -03

Array3DSum.java

BoundaryMatrix.java

DiagonalSum.java

ElementFrequency.java

JaggedArray.java

LeftRotate.java

Maximum3D.java

PositiveNegative3D.java

Q11.java

Q12.java

Q13.java

Q14.java

RemoveDuplicates.java

Search2D.java

Search3D.java

SecondLargest.java

Outline

Timeline

Servers

Java Projects

TriangularMatrix.java

BoundaryMatrix.java

Search2D.java

SortJaggedArray.java

SparseMatrix.java

BoundaryMatrix.java

BoundaryMatrix

```
1 public class BoundaryMatrix {
2     public static void main(String[] args) {
12
13         for (int i = 0; i < 4; i++) {
14             for (int j = 0; j < 4; j++) {
15
16                 if (i == 0 || i == 3 || j == 0 || j == 3) {
17                     System.out.print(matrix[i][j] + " ");
18                 } else {
19                     System.out.print(s; " ");
20                 }
21             }
22             System.out.println();
23         }
24     }
25 }
```

Problems

Output

Debug Console

Terminal

Ports

PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-p
review' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\work
spaceStorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'Bou
ndaryMatrix'
Boundary elements:
1 2 3 4
5 8
9 12
13 14 15 16
PS C:\Users\HP-PC\java lab sheet -03>

Run: LeftRo...

Run: Eleme...

Run: Secon...

Run: Remo...

Run: Diago...

Run: Triang...

Run: ...

Run: Searc...

Run: SortJa...

Run: Spars...

Chat

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

Dictation

Speak and it becomes text. Adjust settings or how it's written any time.

Listening...

Agent Models

Local Default permissions

Ln 25, Col 2 Spaces: 4 UTF-8 CRLF () Java Go Live

Explorer

Open Editors

StudentMarks.java

LeftRotate.java

ElementFrequency.java

SecondLargest.java

RemoveDuplicates.java

DiagonalSum.java

TriangularMatrix.java

BoundaryMatrix.java

Search2D.java

java lab sheet -03

Array3DSum.java

BoundaryMatrix.java

DiagonalSum.java

ElementFrequency.java

JaggedArray.java

LeftRotate.java

Maximum3D.java

PositiveNegative3D.java

Q11.java

Q12.java

Q13.java

Q14.java

RemoveDuplicates.java

Search2D.java

Search3D.java

SecondLargest.java

Outline

Timeline

Servers

Java Projects

TriangularMatrix.java

BoundaryMatrix.java

Search2D.java

SortJaggedArray.java

SparseMatrix.java

Search2D

1 public class Search2D {

2 public static void main(String[] args) {

3

4 int[][] arr = {

5 {10, 20, 30},

6 {40, 50, 60},

7 {70, 80, 90}

8 };

9

10 int search = 50;

11 boolean found = false;

12

13 for (int i = 0; i < arr.length; i++) {

14 for (int j = 0; j < arr[i].length; j++) {

15

16 if (arr[i][j] == search) {

17 System.out.println("Element found at row "

18 + (i + 1) + " and column " + (j + 1));

19 found = true;

20 }

21 }

22 }

Problems

Output

Debug Console

Terminal

Ports

PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\work spaceStorage\91f7735a9b53cbfaa14eeFa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'Search2D'

Element found at row 2 and column 2

PS C:\Users\HP-PC\java lab sheet -03>

Run: LeftRo...

Run: Eleme...

Run: Secon...

Run: Remo...

Run: Diago...

Run: Triang...

Run: Boun...

Run: ...

Run: SortJa...

Run: Spars...

Chat

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

Dictation

Speak and it becomes text. Adjust settings or how it's written any time.

Listening...

Search2D.java

Agent Models

Local Default permissions

Ln 28, Col 2 Spaces: 4 UTF-8 CRLF {} Java Go Live

Explorer

Open Editors

StudentMarks.java

LeftRotate.java

ElementFrequency.java

SecondLargest.java

RemoveDuplicates.java

DiagonalSum.java

TriangularMatrix.java

BoundaryMatrix.java

Search2D.java

java lab sheet -03

Array3DSum.java

BoundaryMatrix.java

DiagonalSum.java

ElementFrequency.java

JaggedArray.java

LeftRotate.java

Maximum3D.java

PositiveNegative3D.java

Q11.java

Q12.java

Q13.java

Q14.java

RemoveDuplicates.java

Search2D.java

Search3D.java

SecondLargest.java

Outline

Timeline

Servers

Java Projects

TriangularMatrix.java

BoundaryMatrix.java

Search2D.java

SortJaggedArray.java

SparseMatrix.java

Search2D

1 public class Search2D {

2 public static void main(String[] args) {

13 for (int i = 0; i < arr.length; i++) {

14 for (int j = 0; j < arr[i].length; j++) {

15

16 if (arr[i][j] == search) {

17 System.out.println("Element found at row "

18 + (i + 1) + " and column " + (j + 1));

19 found = true;

20 }

21 }

22 }

23

24 if (!found) {

25 System.out.println(x: "Element not found");

26 }

27 }

28 }

Problems

Output

Debug Console

Terminal

Ports

PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-p

review' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\work

spacestorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'sea

rch2D'

Element found at row 2 and column 2

PS C:\Users\HP-PC\java lab sheet -03>

Run: LeftRo...

Run: Eleme...

Run: Secon...

Run: Remo...

Run: Diago...

Run: Triang...

Run: Boun...

Run: ...

Run: SortJa...

Run: Spars...

Chat

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI

onto your codebase.

Dictation

Speak and it becomes text. Adjust

settings or how it's written any time.

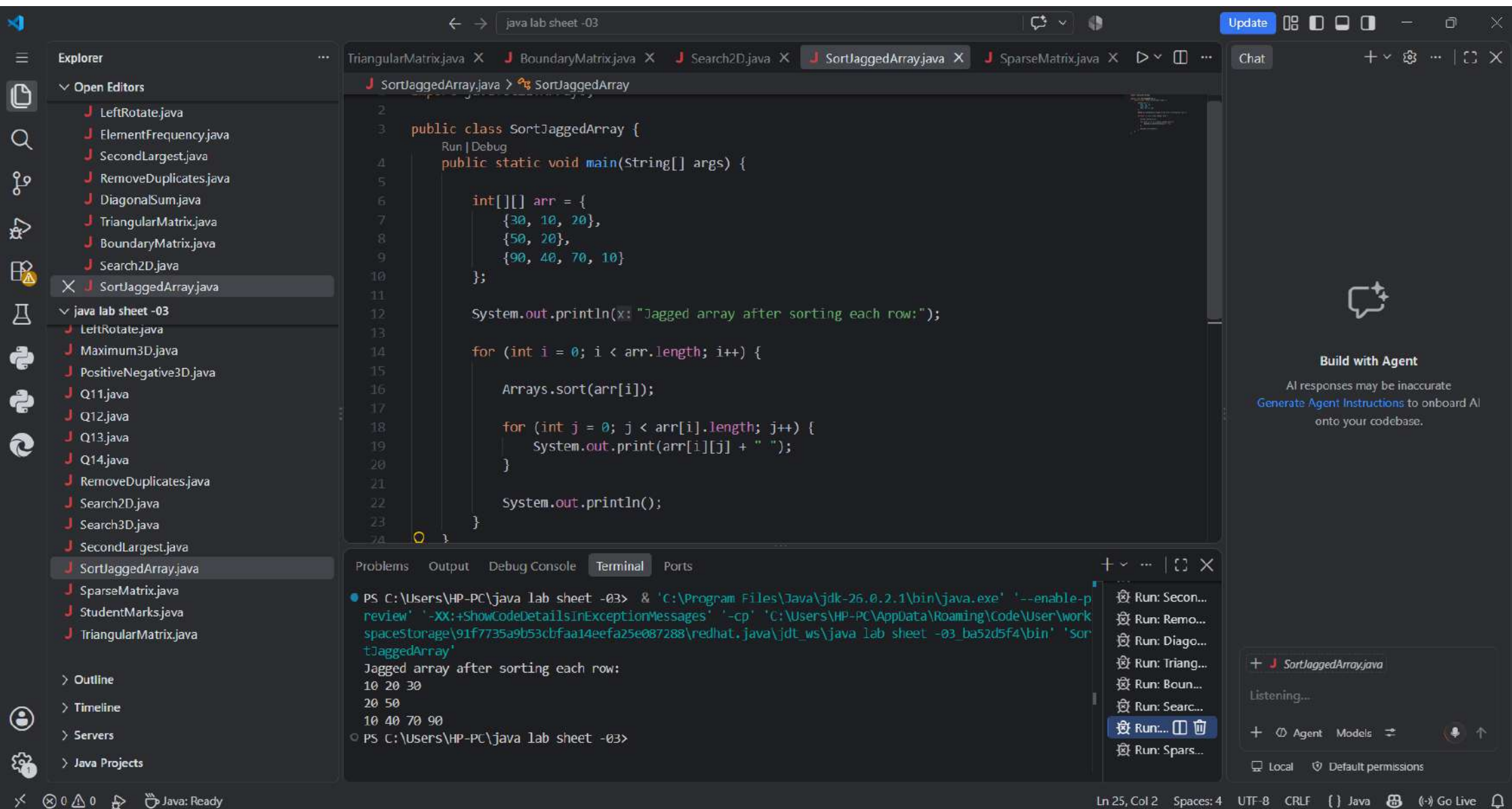
+ Search2D.java

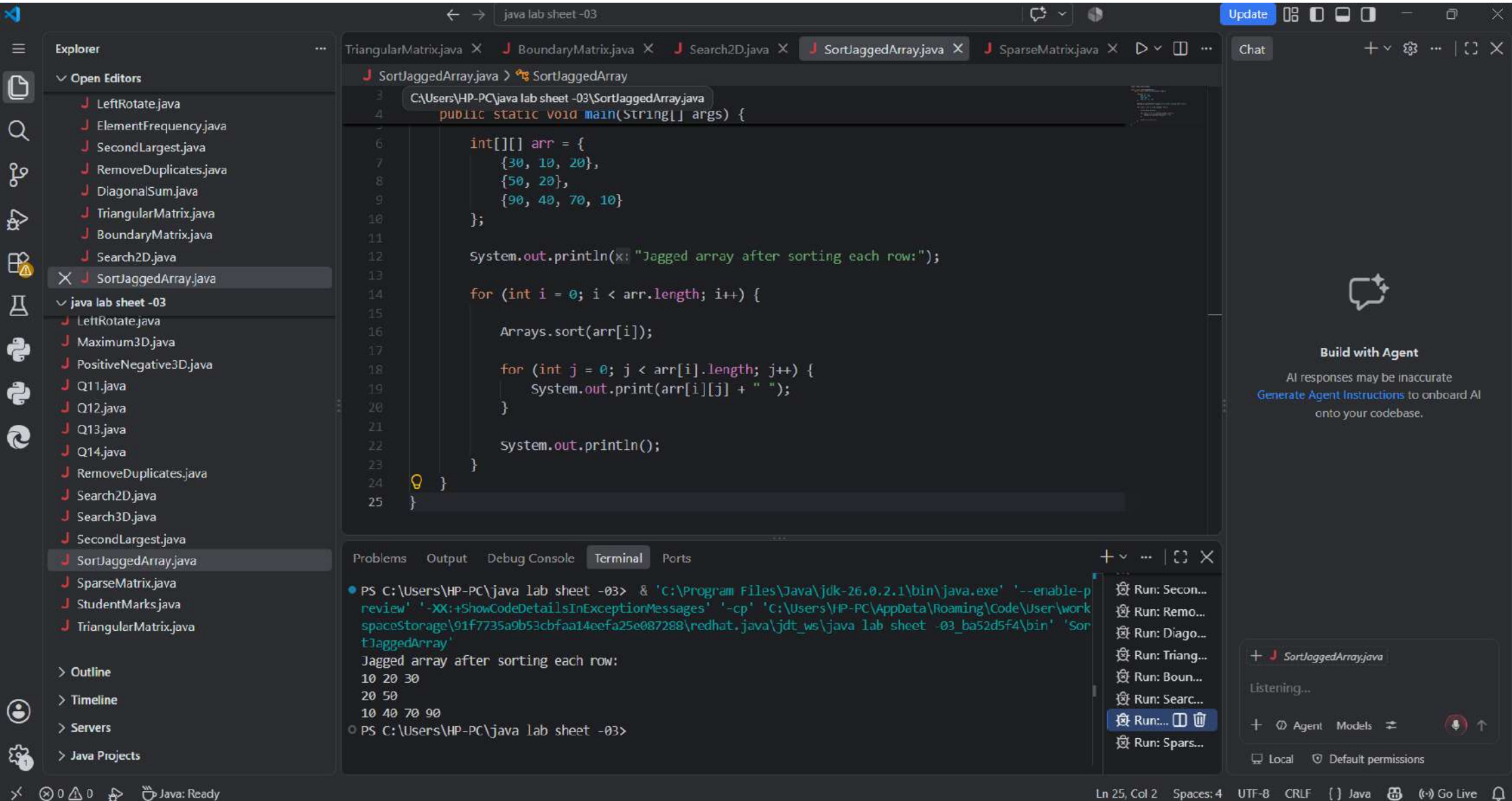
Listening...

+ Agent Models

Local Default permissions

Ln 28, Col 2 Spaces: 4 UTF-8 CRLF {} Java Go Live





Explorer

Open Editors

ElementFrequency.java

SecondLargest.java

RemoveDuplicates.java

DiagonalSum.java

TriangularMatrix.java

BoundaryMatrix.java

Search2D.java

SortJaggedArray.java

X

SparseMatrix.java

java lab sheet -03

LeftRotate.java

Maximum3D.java

PositiveNegative3D.java

Q11.java

Q12.java

Q13.java

Q14.java

RemoveDuplicates.java

Search2D.java

Search3D.java

SecondLargest.java

SortJaggedArray.java

J

SparseMatrix.java

StudentMarks.java

TriangularMatrix.java

Outline

Timeline

Servers

Java Projects

TriangularMatrix.java

BoundaryMatrix.java

Search2D.java

SortJaggedArray.java

J

SparseMatrix.java

J

SparseMatrix.java

Run | Debug

```
1 public class SparseMatrix {
2     public static void main(String[] args) {
3
4         int[][] matrix = {
5             {0, 0, 3},
6             {0, 0, 0},
7             {4, 0, 0}
8         };
9
10        int zero = 0;
11        int nonZero = 0;
12
13        for (int i = 0; i < matrix.length; i++) {
14            for (int j = 0; j < matrix[i].length; j++) {
15
16                if (matrix[i][j] == 0) {
17                    zero++;
18                } else {
19                    nonZero++;
20                }
21            }
22        }
23    }
24 }
```

Problems

Output

Debug Console

Terminal

Ports

PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\workspaceStorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'SparseMatrix'

Number of zero elements = 7

Number of non-zero elements = 2

The matrix is a sparse matrix.

PS C:\Users\HP-PC\java lab sheet -03>

Run: Secon...

Run: Remo...

Run: Diago...

Run: Triang...

Run: Boun...

Run: Searc...

Run: SortJa...

Run:...

Chat

Build with Agent

AI responses may be inaccurate

Generate Agent instructions to onboard AI onto your codebase.

+ J SparseMatrix.java

Listening...

+ Agent Models

Local

Default permissions

Ln 33, Col 2

Spaces: 4

UTF-8

CRLF

{ }

Java

Go Live

Explorer

Open Editors

ElementFrequency.java

SecondLargest.java

RemoveDuplicates.java

DiagonalSum.java

TriangularMatrix.java

BoundaryMatrix.java

Search2D.java

SortJaggedArray.java

X

SparseMatrix.java

java lab sheet -03

LeftRotate.java

Maximum3D.java

PositiveNegative3D.java

Q11.java

Q12.java

Q13.java

Q14.java

RemoveDuplicates.java

Search2D.java

Search3D.java

SecondLargest.java

SortJaggedArray.java

J

SparseMatrix.java

StudentMarks.java

TriangularMatrix.java

Outline

Timeline

Servers

Java Projects

TriangularMatrix.java

J

BoundaryMatrix.java

J

Search2D.java

J

SortJaggedArray.java

J

SparseMatrix.java

J

SparseMatrix

1 public class SparseMatrix {

2 public static void main(String[] args) {

15

16 if (matrix[i][j] == 0) {

17 zero++;

18 } else {

19 nonZero++;

20 }

21 }

22 }

23

24 System.out.println("Number of zero elements = " + zero);

25 System.out.println("Number of non-zero elements = " + nonZero);

26

27 if (zero > nonZero) {

28 System.out.println(x: "The matrix is a sparse matrix.");

29 } else {

30 System.out.println(x: "The matrix is not a sparse matrix.");

31 }

32 }

33 }

Problems

Output

Debug Console

Terminal

Ports

PS C:\Users\HP-PC\java lab sheet -03> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\workspaceStorage\91f7735a9b53cbfaa14eefa25e087288\redhat.java\jdt_ws\java lab sheet -03_ba52d5f4\bin' 'SparseMatrix'

Number of zero elements = 7

Number of non-zero elements = 2

The matrix is a sparse matrix.

PS C:\Users\HP-PC\java lab sheet -03>

Run: Secon...

Run: Remo...

Run: Diago...

Run: Triang...

Run: Boun...

Run: Searc...

Run: SortJa...

Run: ...

Chat

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

+ SparseMatrix.java

Listening...

+ Agent Models

Local Default permissions

Ln 33, Col 2 Spaces: 4 UTF-8 CRLF {} Java Go Live